v7 multi-model matched-notice reproducibility supplement

Research review copy

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1 v7 reproducibility and exploratory statistical audit

The fixed synthesis plan, original source manifest, model registry, software versions, code, source CSV snapshots and validation receipts accompany this appendix. The original raw outputs and manuscript snapshot are preserved. No model was rerun for the synthesis. DiscrimEval input and saved activation hashes were verified for all 21,000 rows. KoBBQ preserves scalar readouts and rendered inputs but does not retain every trial's full hidden state. The historical Qwen axis binary cannot be completely linked to its original creation-time execution. A copied Llama 3.2 planning description contained a DeepSeek metadata URL and layer-15 text; actual configuration, extraction and saved layer-14 axes establish the executed setting. Original planning files were not overwritten.

For item i, let $d_i = 100(p_{i,T} - p_{i,C})$ for a probability outcome. The effect is $n^{-1}\sum_i d_i$. The conditional stereotype measure is computed item by item as $s_i = p_{stereo,i}/(p_{stereo,i} + p_{anti,i})$ before averaging. For DiscrimEval, $m_i = \log\sum_{yes,Yes}\exp(l_{it}) - \log\sum_{no,No}\exp(l_{it})$ and $q_i = \sigma(m_i)$. Thus $m_i = 0$ gives $q_i = 0.5$, while $m_i = 1$ gives approximately 0.731. Mean q and mean m answer different questions. For example, Qwen withholding probabilities of approximately 84.012 and 76.335 percent give a matched effect of 7.677 percentage points. Direct cue interactions are $(T_V - C_V) - (T - C)$; direct model differences subtract effects on identical cluster keys.

All p-values below are post hoc. Sign-flip p-values use a plus-one correction and sign symmetry, not randomized prompt assignment. Each statistical family is fully reported, including nonsignificant results. Bootstrap intervals are individual percentile intervals and do not become simultaneous intervals after p-value adjustment. Cluster leave-one-out ranges and category omissions are sensitivity summaries, not new independent replications. Correlations use cluster-level paired changes and do not identify causal mediation. No demographic equivalence or deployment fairness conclusion follows from nonsignificance.

1.1 All within-model behavioral effects

task	model	metric	contrast	est	lo	hi	p_holm
KoBBQ	Qwen	p_unknown	main	7.677	3.139	12.562	0.027
KoBBQ	Qwen	p_unknown	V	7.700	3.709	12.164	0.00435
KoBBQ	Qwen	p_sgp	main	-1.972	-7.125	3.250	1
KoBBQ	Qwen	p_sgp	V	-0.967	-5.815	3.936	1
KoBBQ	Llama31	p_unknown	main	-8.296	-10.818	-5.829	0.0018
KoBBQ	Llama31	p_unknown	V	-0.408	-2.753	2.059	1
KoBBQ	Llama31	p_sgp	main	-0.661	-2.843	1.492	1
KoBBQ	Llama31	p_sgp	V	-0.286	-2.846	2.205	1
KoBBQ	DeepSeek	p_unknown	main	2.360	0.482	4.072	0.122
KoBBQ	DeepSeek	p_unknown	V	-0.696	-2.906	1.250	1
KoBBQ	DeepSeek	p_sgp	main	-2.015	-4.727	0.446	1
KoBBQ	DeepSeek	p_sgp	V	0.219	-2.933	3.335	1
KoBBQ	Gemma2	p_unknown	main	21.069	14.635	28.235	0.0018
KoBBQ	Gemma2	p_unknown	V	25.333	18.856	32.344	0.0018
KoBBQ	Gemma2	p_sgp	main	0.353	-3.511	4.181	1
KoBBQ	Gemma2	p_sgp	V	0.273	-3.596	4.214	1
KoBBQ	Llama32	p_unknown	main	6.398	4.727	8.131	0.0018
KoBBQ	Llama32	p_unknown	V	11.353	9.606	13.074	0.0018
KoBBQ	Llama32	p_sgp	main	-1.409	-4.140	1.376	1
KoBBQ	Llama32	p_sgp	V	-0.879	-3.262	1.419	1
DiscrimEval	Qwen	q_yes	main	5.433	2.308	9.190	0.0196
DiscrimEval	Qwen	q_yes	V	-6.159	-10.272	-2.164	0.0638
DiscrimEval	Qwen	margin	main	0.166	-0.373	0.722	1
DiscrimEval	Qwen	margin	V	-2.730	-3.415	-2.052	0.0018
DiscrimEval	DeepSeek	q_yes	main	2.169	0.759	3.753	0.11
DiscrimEval	DeepSeek	q_yes	V	-1.738	-3.475	0.165	1
DiscrimEval	DeepSeek	margin	main	0.022	-0.095	0.143	1
DiscrimEval	DeepSeek	margin	V	-0.595	-0.807	-0.373	0.0018

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task	model	metric	contrast	est	lo	hi	p_holm
DiscrimEval	Gemma2	q_yes	main	0.937	-3.574	5.417	1
DiscrimEval	Gemma2	q_yes	V	5.664	0.439	10.687	0.673
DiscrimEval	Gemma2	margin	main	-0.039	-0.411	0.325	1
DiscrimEval	Gemma2	margin	V	0.336	-0.037	0.705	1
DiscrimEval	Llama32	q_yes	main	-0.897	-2.625	0.960	1
DiscrimEval	Llama32	q_yes	V	-1.355	-2.982	0.348	1
DiscrimEval	Llama32	margin	main	-0.292	-0.481	-0.107	0.0559
DiscrimEval	Llama32	margin	V	-0.251	-0.427	-0.084	0.11

1.2 Cue interactions

task	model	metric	est	lo	hi	p_holm
KoBBQ	Qwen	p_unknown	0.023	-3.029	2.760	1
KoBBQ	Qwen	p_sgp	1.005	-2.162	4.165	1
KoBBQ	Llama31	p_unknown	7.888	6.186	9.780	0.0009
KoBBQ	Llama31	p_sgp	0.375	-1.040	1.761	1
KoBBQ	DeepSeek	p_unknown	-3.056	-5.280	-1.128	0.014
KoBBQ	DeepSeek	p_sgp	2.234	-0.631	5.085	1
KoBBQ	Gemma2	p_unknown	4.263	0.257	8.178	0.367
KoBBQ	Gemma2	p_sgp	-0.079	-2.026	1.773	1
KoBBQ	Llama32	p_unknown	4.955	3.554	6.433	0.0009
KoBBQ	Llama32	p_sgp	0.531	-1.108	2.153	1
DiscrimEval	Qwen	q_yes	-11.592	-15.313	-8.245	0.0009
DiscrimEval	Qwen	margin	-2.896	-3.277	-2.512	0.0009
DiscrimEval	DeepSeek	q_yes	-3.907	-4.839	-3.039	0.0009
DiscrimEval	DeepSeek	margin	-0.617	-0.746	-0.488	0.0009
DiscrimEval	Gemma2	q_yes	4.727	2.359	7.284	0.00275
DiscrimEval	Gemma2	margin	0.375	0.215	0.542	0.0012
DiscrimEval	Llama32	q_yes	-0.458	-1.511	0.521	1
DiscrimEval	Llama32	margin	0.041	-0.032	0.113	1

1.3 Direct model comparisons in KoBBQ

model_a	model_b	metric	contrast	est	lo	hi	p_holm
Qwen	Llama31	p_unknown	main	15.972	10.515	21.460	0.0048
Qwen	Llama31	p_unknown	V	8.108	3.527	13.022	0.0504
Qwen	Llama31	p_unknown	V_minus_main	-7.864	-11.227	-4.638	0.0048
Qwen	Llama31	p_sgp	main	-1.311	-7.545	4.964	1
Qwen	Llama31	p_sgp	V	-0.681	-7.175	6.127	1
Qwen	Llama31	p_sgp	V_minus_main	0.630	-2.902	4.301	1
Qwen	DeepSeek	p_unknown	main	5.317	0.621	10.277	1
Qwen	DeepSeek	p_unknown	V	8.396	3.843	13.294	0.017
Qwen	DeepSeek	p_unknown	V_minus_main	3.079	-0.780	6.741	1
Qwen	DeepSeek	p_sgp	main	0.043	-5.621	5.659	1
Qwen	DeepSeek	p_sgp	V	-1.186	-7.257	5.064	1
Qwen	DeepSeek	p_sgp	V_minus_main	-1.229	-5.936	3.276	1
Qwen	Gemma2	p_unknown	main	-13.392	-21.881	-5.118	0.107
Qwen	Gemma2	p_unknown	V	-17.633	-25.190	-10.129	0.0048
Qwen	Gemma2	p_unknown	V_minus_main	-4.240	-9.517	0.596	1
Qwen	Gemma2	p_sgp	main	-2.325	-9.059	4.585	1
Qwen	Gemma2	p_sgp	V	-1.240	-7.581	5.134	1
Qwen	Gemma2	p_sgp	V_minus_main	1.084	-2.509	4.751	1
Qwen	Llama32	p_unknown	main	1.279	-3.675	6.422	1
Qwen	Llama32	p_unknown	V	-3.653	-7.788	0.893	1
Qwen	Llama32	p_unknown	V_minus_main	-4.932	-8.228	-1.902	0.0793
Qwen	Llama32	p_sgp	main	-0.563	-5.875	4.785	1
Qwen	Llama32	p_sgp	V	-0.088	-5.006	5.083	1
Qwen	Llama32	p_sgp	V_minus_main	0.474	-3.035	4.106	1
Llama31	DeepSeek	p_unknown	main	-10.656	-13.840	-7.625	0.0048
Llama31	DeepSeek	p_unknown	V	0.288	-2.535	3.032	1
Llama31	DeepSeek	p_unknown	V_minus_main	10.944	8.184	13.874	0.0048
Llama31	DeepSeek	p_sgp	main	1.354	-2.075	4.623	1
Llama31	DeepSeek	p_sgp	V	-0.505	-4.246	3.093	1
Llama31	DeepSeek	p_sgp	V_minus_main	-1.859	-4.810	1.120	1
Llama31	Gemma2	p_unknown	main	-29.365	-36.836	-22.623	0.0048
Llama31	Gemma2	p_unknown	V	-25.741	-32.607	-19.410	0.0048
Llama31	Gemma2	p_unknown	V_minus_main	3.624	-0.657	7.907	1
Llama31	Gemma2	p_sgp	main	-1.014	-5.223	3.217	1
Llama31	Gemma2	p_sgp	V	-0.560	-4.755	3.800	1
Llama31	Gemma2	p_sgp	V_minus_main	0.454	-1.949	2.834	1
Llama31	Llama32	p_unknown	main	-14.693	-17.354	-12.037	0.0048
Llama31	Llama32	p_unknown	V	-11.761	-14.457	-8.851	0.0048
Llama31	Llama32	p_unknown	V_minus_main	2.932	0.974	5.065	0.308
Llama31	Llama32	p_sgp	main	0.748	-3.401	4.738	1
Llama31	Llama32	p_sgp	V	0.593	-2.901	4.067	1
Llama31	Llama32	p_sgp	V_minus_main	-0.156	-1.828	1.528	1

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model_a	model_b	metric	contrast	est	lo	hi	p_holm
DeepSeek	Gemma2	p_unknown	main	-18.709	-25.934	-11.865	0.0048
DeepSeek	Gemma2	p_unknown	V	-26.029	-33.050	-19.415	0.0048
DeepSeek	Gemma2	p_unknown	V_minus_main	-7.319	-11.600	-2.911	0.0682
DeepSeek	Gemma2	p_sgp	main	-2.368	-7.217	2.447	1
DeepSeek	Gemma2	p_sgp	V	-0.054	-4.855	4.712	1
DeepSeek	Gemma2	p_sgp	V_minus_main	2.313	-1.178	5.690	1
DeepSeek	Llama32	p_unknown	main	-4.038	-6.588	-1.460	0.106
DeepSeek	Llama32	p_unknown	V	-12.049	-14.899	-9.281	0.0048
DeepSeek	Llama32	p_unknown	V_minus_main	-8.011	-10.511	-5.508	0.0048
DeepSeek	Llama32	p_sgp	main	-0.606	-4.658	3.092	1
DeepSeek	Llama32	p_sgp	V	1.098	-2.843	5.020	1
DeepSeek	Llama32	p_sgp	V_minus_main	1.703	-1.394	4.874	1
Gemma2	Llama32	p_unknown	main	14.672	7.715	22.170	0.0105
Gemma2	Llama32	p_unknown	V	13.980	6.890	21.527	0.0325
Gemma2	Llama32	p_unknown	V_minus_main	-0.692	-4.777	3.584	1
Gemma2	Llama32	p_sgp	main	1.762	-2.569	5.902	1
Gemma2	Llama32	p_sgp	V	1.152	-2.847	4.864	1
Gemma2	Llama32	p_sgp	V_minus_main	-0.610	-3.087	1.866	1

1.4 Direct model comparisons in DiscrimEval

model_a	model_b	metric	contrast	est	lo	hi	p_holm
Qwen	DeepSeek	q_yes	main	3.264	-0.471	7.426	1
Qwen	DeepSeek	q_yes	V	-4.421	-8.928	0.034	1
Qwen	DeepSeek	q_yes	V_minus_main	-7.685	-11.405	-4.310	0.0048
Qwen	DeepSeek	margin	main	0.144	-0.426	0.722	1
Qwen	DeepSeek	margin	V	-2.135	-2.785	-1.469	0.0048
Qwen	DeepSeek	margin	V_minus_main	-2.279	-2.629	-1.931	0.0048
Qwen	Gemma2	q_yes	main	4.496	-0.605	9.657	1
Qwen	Gemma2	q_yes	V	-11.823	-17.543	-6.301	0.017
Qwen	Gemma2	q_yes	V_minus_main	-16.319	-21.027	-11.982	0.0048
Qwen	Gemma2	margin	main	0.205	-0.479	0.875	1
Qwen	Gemma2	margin	V	-3.065	-3.837	-2.321	0.0048
Qwen	Gemma2	margin	V_minus_main	-3.271	-3.663	-2.870	0.0048
Qwen	Llama32	q_yes	main	6.330	2.630	10.417	0.0793
Qwen	Llama32	q_yes	V	-4.804	-9.474	-0.308	1
Qwen	Llama32	q_yes	V_minus_main	-11.134	-14.666	-7.961	0.0048
Qwen	Llama32	margin	main	0.458	-0.132	1.044	1
Qwen	Llama32	margin	V	-2.478	-3.205	-1.766	0.0048
Qwen	Llama32	margin	V_minus_main	-2.937	-3.337	-2.544	0.0048
DeepSeek	Gemma2	q_yes	main	1.232	-3.447	6.018	1
DeepSeek	Gemma2	q_yes	V	-7.402	-12.900	-2.023	0.416
DeepSeek	Gemma2	q_yes	V_minus_main	-8.634	-11.557	-6.045	0.0048
DeepSeek	Gemma2	margin	main	0.061	-0.347	0.472	1
DeepSeek	Gemma2	margin	V	-0.930	-1.408	-0.445	0.0264
DeepSeek	Gemma2	margin	V_minus_main	-0.992	-1.203	-0.772	0.0048
DeepSeek	Llama32	q_yes	main	3.066	0.855	5.381	0.392
DeepSeek	Llama32	q_yes	V	-0.383	-2.284	1.622	1
DeepSeek	Llama32	q_yes	V_minus_main	-3.449	-4.880	-1.906	0.0048
DeepSeek	Llama32	margin	main	0.314	0.122	0.513	0.108
DeepSeek	Llama32	margin	V	-0.343	-0.580	-0.093	0.349
DeepSeek	Llama32	margin	V_minus_main	-0.658	-0.828	-0.498	0.0048
Gemma2	Llama32	q_yes	main	1.834	-2.794	6.577	1
Gemma2	Llama32	q_yes	V	7.019	1.581	12.524	0.66
Gemma2	Llama32	q_yes	V_minus_main	5.185	2.619	7.958	0.0138
Gemma2	Llama32	margin	main	0.253	-0.173	0.671	1
Gemma2	Llama32	margin	V	0.587	0.142	1.031	0.606
Gemma2	Llama32	margin	V_minus_main	0.334	0.153	0.524	0.0416

1.5 Demographic gap changes for q yes

model	group	contrast	record_gap	threat_gap	est	p_holm
Qwen	Asian	main	2.372	2.214	-0.158	1
Qwen	Asian	V	2.206	1.679	-0.527	1
Qwen	Black	main	3.275	4.379	1.104	1
Qwen	Black	V	3.420	4.068	0.648	1
Qwen	Hispanic	main	2.297	4.536	2.238	1
Qwen	Hispanic	V	1.838	3.851	2.012	1
Qwen	Native American	main	4.305	4.115	-0.190	1
Qwen	Native American	V	4.212	3.681	-0.531	1
Qwen	male	main	-1.258	-0.958	0.301	1
Qwen	male	V	-1.397	-1.116	0.281	1
Qwen	non-binary	main	1.346	3.093	1.747	1
Qwen	non-binary	V	0.919	2.776	1.858	1
DeepSeek	Asian	main	0.575	0.108	-0.467	0.45
DeepSeek	Asian	V	0.571	0.226	-0.345	1

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model	group	contrast	record_gap	threat_gap	est	p_holm
DeepSeek	Black	main	1.426	0.509	-0.918	0.0093
DeepSeek	Black	V	1.377	0.839	-0.538	1
DeepSeek	Hispanic	main	1.037	0.187	-0.850	0.0522
DeepSeek	Hispanic	V	0.884	0.767	-0.117	1
DeepSeek	Native American	main	1.042	0.179	-0.863	0.172
DeepSeek	Native American	V	0.913	0.475	-0.437	1
DeepSeek	male	main	-0.449	-0.191	0.259	1
DeepSeek	male	V	-0.543	-0.261	0.281	1
DeepSeek	non-binary	main	0.190	0.362	0.172	1
DeepSeek	non-binary	V	-0.171	0.017	0.188	1
Gemma2	Asian	main	1.307	1.175	-0.132	1
Gemma2	Asian	V	1.045	2.190	1.145	1
Gemma2	Black	main	2.535	4.251	1.716	1
Gemma2	Black	V	2.563	4.381	1.818	0.967
Gemma2	Hispanic	main	1.187	1.292	0.105	1
Gemma2	Hispanic	V	1.144	2.491	1.347	1
Gemma2	Native American	main	1.830	2.240	0.410	1
Gemma2	Native American	V	1.749	3.168	1.420	1
Gemma2	male	main	-1.298	-1.671	-0.373	1
Gemma2	male	V	-1.161	-1.632	-0.471	1
Gemma2	non-binary	main	2.883	2.762	-0.121	1
Gemma2	non-binary	V	2.878	3.333	0.454	1
Llama32	Asian	main	0.581	0.766	0.184	1
Llama32	Asian	V	0.572	0.815	0.243	1
Llama32	Black	main	1.541	2.377	0.837	0.223
Llama32	Black	V	1.543	2.055	0.513	1
Llama32	Hispanic	main	0.661	1.474	0.813	0.0356
Llama32	Hispanic	V	0.499	1.339	0.840	0.0182
Llama32	Native American	main	0.772	1.302	0.530	1
Llama32	Native American	V	0.520	1.396	0.876	0.265
Llama32	male	main	-0.392	-0.911	-0.518	1
Llama32	male	V	-0.551	-0.799	-0.249	1
Llama32	non-binary	main	1.349	2.339	0.990	0.533
Llama32	non-binary	V	1.356	2.199	0.843	1

1.6 Demographic gap changes for margin

model	group	contrast	record_gap	threat_gap	est	p_holm
Qwen	Asian	main	0.363	0.410	0.047	1
Qwen	Asian	V	0.438	0.382	-0.056	1
Qwen	Black	main	0.788	0.855	0.067	1
Qwen	Black	V	0.764	0.826	0.062	1
Qwen	Hispanic	main	0.558	0.668	0.110	1
Qwen	Hispanic	V	0.557	0.641	0.084	1
Qwen	Native American	main	0.667	0.606	-0.060	1
Qwen	Native American	V	0.695	0.625	-0.070	1
Qwen	male	main	-0.259	-0.239	0.020	1
Qwen	male	V	-0.270	-0.160	0.110	1
Qwen	non-binary	main	0.356	0.479	0.122	1
Qwen	non-binary	V	0.316	0.624	0.309	1
DeepSeek	Asian	main	0.052	0.020	-0.032	1
DeepSeek	Asian	V	0.054	0.010	-0.044	1
DeepSeek	Black	main	0.146	0.074	-0.072	0.0138
DeepSeek	Black	V	0.143	0.074	-0.069	0.0048
DeepSeek	Hispanic	main	0.105	0.054	-0.050	0.736
DeepSeek	Hispanic	V	0.097	0.061	-0.036	1
DeepSeek	Native American	main	0.079	0.040	-0.039	1
DeepSeek	Native American	V	0.081	0.033	-0.049	1
DeepSeek	male	main	-0.035	-0.010	0.025	1
DeepSeek	male	V	-0.032	-0.025	0.007	1
DeepSeek	non-binary	main	-0.008	-0.013	-0.005	1
DeepSeek	non-binary	V	-0.029	-0.041	-0.011	1
Gemma2	Asian	main	0.151	0.167	0.015	1
Gemma2	Asian	V	0.133	0.204	0.071	1
Gemma2	Black	main	0.233	0.433	0.200	0.0048
Gemma2	Black	V	0.227	0.421	0.194	0.0048
Gemma2	Hispanic	main	0.140	0.184	0.045	1
Gemma2	Hispanic	V	0.136	0.229	0.093	0.433
Gemma2	Native American	main	0.154	0.269	0.116	0.166
Gemma2	Native American	V	0.159	0.294	0.135	0.0182
Gemma2	male	main	-0.096	-0.162	-0.065	1
Gemma2	male	V	-0.083	-0.173	-0.090	1
Gemma2	non-binary	main	0.245	0.296	0.051	1
Gemma2	non-binary	V	0.251	0.331	0.080	1
Llama32	Asian	main	0.117	0.129	0.012	1
Llama32	Asian	V	0.113	0.116	0.003	1
Llama32	Black	main	0.195	0.247	0.053	0.0731
Llama32	Black	V	0.179	0.220	0.041	1
Llama32	Hispanic	main	0.153	0.216	0.063	0.0808
Llama32	Hispanic	V	0.128	0.193	0.065	0.0356
Llama32	Native American	main	0.174	0.205	0.031	1

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model	group	contrast	record_gap	threat_gap	est	p_holm
Llama32	Native American	V	0.139	0.197	0.058	0.483
Llama32	male	main	-0.070	-0.131	-0.061	0.26
Llama32	male	V	-0.063	-0.117	-0.054	0.0966
Llama32	non-binary	main	0.229	0.255	0.026	1
Llama32	non-binary	V	0.219	0.242	0.022	1

1.7 Emotion associations in KoBBQ

model	contrast	axis	outcome	rho	p_bh
Qwen	main	Fear	p_unknown	-0.003	0.987
Qwen	main	Fear	p_sgp	0.073	0.96
Qwen	main	Anxiety	p_unknown	-0.062	0.96
Qwen	main	Anxiety	p_sgp	0.090	0.96
Qwen	main	Certainty	p_unknown	0.038	0.96
Qwen	main	Certainty	p_sgp	-0.010	0.96
Qwen	V	Fear	p_unknown	0.028	0.96
Qwen	V	Fear	p_sgp	-0.023	0.96
Qwen	V	Anxiety	p_unknown	-0.012	0.96
Qwen	V	Anxiety	p_sgp	0.004	0.987
Qwen	V	Certainty	p_unknown	-0.076	0.96
Qwen	V	Certainty	p_sgp	-0.028	0.96
Llama31	main	Fear	p_unknown	0.189	0.451
Llama31	main	Fear	p_sgp	0.075	0.96
Llama31	main	Anxiety	p_unknown	0.219	0.249
Llama31	main	Anxiety	p_sgp	0.031	0.96
Llama31	main	Certainty	p_unknown	-0.430	0.000305
Llama31	main	Certainty	p_sgp	-0.020	0.96
Llama31	V	Fear	p_unknown	0.038	0.96
Llama31	V	Fear	p_sgp	-0.042	0.96
Llama31	V	Anxiety	p_unknown	0.078	0.96
Llama31	V	Anxiety	p_sgp	-0.054	0.96
Llama31	V	Certainty	p_unknown	-0.421	0.000305
Llama31	V	Certainty	p_sgp	-0.010	0.96
DeepSeek	main	Fear	p_unknown	-0.008	0.963
DeepSeek	main	Fear	p_sgp	0.101	0.96
DeepSeek	main	Anxiety	p_unknown	-0.055	0.96
DeepSeek	main	Anxiety	p_sgp	-0.020	0.96
DeepSeek	main	Certainty	p_unknown	0.035	0.96
DeepSeek	main	Certainty	p_sgp	0.058	0.96
DeepSeek	V	Fear	p_unknown	0.233	0.206
DeepSeek	V	Fear	p_sgp	-0.113	0.956
DeepSeek	V	Anxiety	p_unknown	0.097	0.96
DeepSeek	V	Anxiety	p_sgp	-0.112	0.956
DeepSeek	V	Certainty	p_unknown	-0.133	0.916
DeepSeek	V	Certainty	p_sgp	-0.016	0.96
Gemma2	main	Fear	p_unknown	-0.117	0.95
Gemma2	main	Fear	p_sgp	-0.065	0.96
Gemma2	main	Anxiety	p_unknown	-0.000	0.996
Gemma2	main	Anxiety	p_sgp	0.015	0.96
Gemma2	main	Certainty	p_unknown	0.309	0.0201
Gemma2	main	Certainty	p_sgp	-0.017	0.96
Gemma2	V	Fear	p_unknown	-0.096	0.96
Gemma2	V	Fear	p_sgp	0.073	0.96
Gemma2	V	Anxiety	p_unknown	0.009	0.96
Gemma2	V	Anxiety	p_sgp	0.079	0.96
Gemma2	V	Certainty	p_unknown	0.123	0.916
Gemma2	V	Certainty	p_sgp	-0.153	0.719
Llama32	main	Fear	p_unknown	-0.128	0.916
Llama32	main	Fear	p_sgp	-0.170	0.564
Llama32	main	Anxiety	p_unknown	-0.040	0.96
Llama32	main	Anxiety	p_sgp	-0.126	0.916
Llama32	main	Certainty	p_unknown	-0.012	0.96
Llama32	main	Certainty	p_sgp	-0.013	0.96
Llama32	V	Fear	p_unknown	-0.040	0.96
Llama32	V	Fear	p_sgp	-0.093	0.96
Llama32	V	Anxiety	p_unknown	-0.054	0.96
Llama32	V	Anxiety	p_sgp	-0.042	0.96
Llama32	V	Certainty	p_unknown	0.097	0.96
Llama32	V	Certainty	p_sgp	0.026	0.96

1.8 Emotion associations in DiscrimEval

model	contrast	axis	outcome	rho	p_bh
Qwen	main	Fear	q_yes	-0.102	0.96
Qwen	main	Fear	margin	-0.044	0.96
Qwen	main	Anxiety	q_yes	-0.058	0.96

Continued on next page

model	contrast	axis	outcome	rho	p_bh
Qwen	main	Anxiety	margin	-0.066	0.96
Qwen	main	Certainty	q_yes	-0.029	0.96
Qwen	main	Certainty	margin	0.082	0.96
Qwen	V	Fear	q_yes	-0.036	0.96
Qwen	V	Fear	margin	-0.060	0.96
Qwen	V	Anxiety	q_yes	-0.080	0.96
Qwen	V	Anxiety	margin	0.067	0.96
Qwen	V	Certainty	q_yes	0.155	0.916
Qwen	V	Certainty	margin	-0.028	0.96
DeepSeek	main	Fear	q_yes	-0.282	0.215
DeepSeek	main	Fear	margin	-0.426	0.00512
DeepSeek	main	Anxiety	q_yes	-0.097	0.96
DeepSeek	main	Anxiety	margin	-0.228	0.483
DeepSeek	main	Certainty	q_yes	0.085	0.96
DeepSeek	main	Certainty	margin	0.208	0.564
DeepSeek	V	Fear	q_yes	-0.161	0.916
DeepSeek	V	Fear	margin	-0.458	0.0024
DeepSeek	V	Anxiety	q_yes	-0.157	0.916
DeepSeek	V	Anxiety	margin	-0.316	0.118
DeepSeek	V	Certainty	q_yes	0.136	0.96
DeepSeek	V	Certainty	margin	0.448	0.00269
Gemma2	main	Fear	q_yes	0.021	0.96
Gemma2	main	Fear	margin	0.033	0.96
Gemma2	main	Anxiety	q_yes	0.099	0.96
Gemma2	main	Anxiety	margin	0.041	0.96
Gemma2	main	Certainty	q_yes	0.073	0.96
Gemma2	main	Certainty	margin	0.034	0.96
Gemma2	V	Fear	q_yes	0.114	0.96
Gemma2	V	Fear	margin	0.059	0.96
Gemma2	V	Anxiety	q_yes	0.110	0.96
Gemma2	V	Anxiety	margin	0.067	0.96
Gemma2	V	Certainty	q_yes	-0.096	0.96
Gemma2	V	Certainty	margin	-0.053	0.96
Llama32	main	Fear	q_yes	0.036	0.96
Llama32	main	Fear	margin	0.064	0.96
Llama32	main	Anxiety	q_yes	-0.027	0.96
Llama32	main	Anxiety	margin	0.027	0.96
Llama32	main	Certainty	q_yes	0.070	0.96
Llama32	main	Certainty	margin	-0.065	0.96
Llama32	V	Fear	q_yes	-0.068	0.96
Llama32	V	Fear	margin	-0.045	0.96
Llama32	V	Anxiety	q_yes	-0.258	0.302
Llama32	V	Anxiety	margin	-0.214	0.564
Llama32	V	Certainty	q_yes	0.148	0.95
Llama32	V	Certainty	margin	0.030	0.96

1.9 Full-vocabulary KoBBQ unknown-token mass diagnostics

model	contrast	est	lo	hi
Qwen	main	7.651	3.113	12.519
Qwen	V	7.686	3.705	12.124
Llama31	main	-9.705	-11.784	-7.630
Llama31	V	-6.346	-8.112	-4.452
DeepSeek	main	2.178	0.290	3.868
DeepSeek	V	-1.324	-3.357	0.527
Gemma2	main	-12.979	-16.696	-9.175
Gemma2	V	31.459	26.716	36.235
Llama32	main	-0.079	-1.474	1.228
Llama32	V	2.239	1.129	3.365

1.10 Full-vocabulary DiscrimEval affirmative-token mass diagnostics

model	contrast	est	lo	hi
Qwen	main	5.292	2.163	9.077
Qwen	V	-6.365	-10.502	-2.339
Qwen	V_minus_main	-11.657	-15.358	-8.321
DeepSeek	main	2.170	0.763	3.762
DeepSeek	V	-1.714	-3.453	0.190
DeepSeek	V_minus_main	-3.884	-4.818	-3.019
Gemma2	main	1.763	-1.654	5.244
Gemma2	V	5.373	1.324	9.258
Gemma2	V_minus_main	3.610	1.845	5.465
Llama32	main	-1.016	-2.734	0.820
Llama32	V	-1.533	-3.158	0.167

Llama32	V_minus_main	-0.517	-1.564	0.456

These supplementary mass diagnostics were expanded after the quality audit. They report individual intervals only and introduce no new p-value or corrected discovery claim. Conditional probabilities, full-vocabulary token masses and free-generation frequencies are distinct quantities.